

Review

AFTI: High-generalized facial template inversion under attribute guidance

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ABSTRACT

Face recognition has brought tremendous convenience to daily life, enabling applications like access control, mobile payment, and criminal identification. However, facial templates stored on servers pose significant privacy risks due to potential security breaches. To underscore this concern, facial template inversion—reconstructing face images from templates—has been widely studied. While recent approaches adopt pixel-wise/perceptual losses and elaborate networks to minimize reconstruction errors, they suffer from limited performance in Type-II attack scenarios and unsatisfied generalization across unseen models. In this work, we propose Attribute-guided Facial Template Inversion (AFTI), a strategy that leverages facial attributes (e.g., eye shape, nose structure, mouth characteristics) to impose explicit semantic constraints, mitigating the ambiguity of template-to-image mapping. We introduce an attribute loss and integrate it into two frameworks: UCNet (a simple deep network) and the state-of-the-art M-StyleGAN3. Experimental results show AFTI enhances existing SOTA methods, achieving higher True Accept Rates under Type-II attacks and improved cross-model generalization.

1. Introduction

In recent years, biometric traits have been widely adopted for identity recognition in real-world applications. Among these traits, facial features gain immense popularity due to their contactless nature, ease of use, and relatively low implementation costs. They have widespread applications in various domains, such as access control (Ibrahim & Zin, 2011), attendance tracking (Rekha & Ramaprasad, 2017), security check on the subway (Abdul-azeez et al., 2019), and mobile payment (Feng et al., 2017). Despite the convenience, face recognition systems also have notable security concerns.

In face recognition systems, a template (generally a feature vector) is extracted from the enrolled face image. This template is closely associated with the individual's identity and inherently contains sensitive information about the person. However, to facilitate identity verification, these templates are often stored in databases without encryption or adequate security measures. If compromised through storage leaks, transmission intercepts, or insider threats, these templates could pose serious privacy risks.

In this paper, we focus on facial template inversion attacks against face recognition systems and propose the high-generalized attribute-guided facial template inversion (AFTI) strategy to reconstruct face images (samples are shown in Fig. 1). Fig. 2 depicts the general diagram of facial template inversion attacks. Typically, an attacker can compromise a face recognition system in three stages if he gains unauthorized access to a facial template. Initially, the attacker uses the target face recogni-

tion system to generate the training pairs, consisting of face images and their corresponding templates. Subsequently, the attacker would train an inverter using these pairs. Ultimately, the attacker would be able to reconstruct the face image from the corresponding facial template. This attack can be named as the *facial template inversion* (Fredrikson et al., 2015) or *face reconstruction* (Mai et al., 2018). The reconstructed image poses significant threats to privacy and even enables the attacker to potentially bypass face recognition systems. Specifically, potential risks would include:

- The attacker can deduce soft biometric information about the individual, such as gender, age, ethnicity, etc., from the reconstructed face image.
- The attacker can search for additional details about the individual, including their name, home address, and other personal information, utilizing search platforms such as Clearview AI¹ or the Baidu search engine.
- Furthermore, the attacker can utilize the reconstructed face image to gain unauthorized access to the face recognition systems where the individual is registered, potentially leading to further harm.

Many studies have delved into the field of facial template inversion (Adler, 2003; Cole et al., 2017; Dong et al., 2021, 2023; Duong et al.,

¹ <https://www.forbes.com/sites/kateoflahertyuk/2020/02/26/clearview-ai-the-company-whose-database-has-amassed-3-billion-photos-hacked>.

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Fig. 1. Sample face images from the LFW dataset and their corresponding images reconstructed from ArcFace templates.

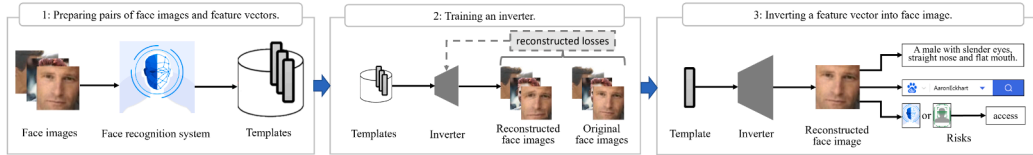


Fig. 2. The general diagram of facial template inversion attack.

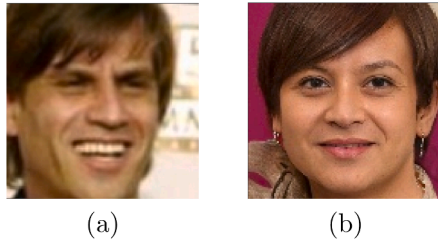


Fig. 3. The reconstructed image can bypass the face recognition system but lacks substantial similarity to the original in a human visual perspective: (a) An original face image from LFW, and (b) The corresponding face image reconstructed by M-StyleGAN3 (Shahreza & Marcel, 2023a).

2020; Fredrikson et al., 2015; Mai et al., 2018; Mignon & Jurie, 2013; Otroshi Shahreza & Marcel, 2024; Shahreza et al., 2024; Shahreza & Marcel, 2023b; Truong et al., 2022; Vendrow & Vendrow, 2021), aiming to reconstruct face images capable of bypassing target recognition systems. However, as shown in Fig. 3, existing methods often fail to preserve critical facial attributes (e.g., eyes, nose, mouth) in reconstructed images, limiting their perceptual fidelity and attack success rates. Generally, the templates in face recognition systems are expected to be highly associated with the facial attributes to better represent the face identity. Reconstructed images with similar facial attributes could have better generalization to bypass face recognition systems.

To address the existing research gaps in this field, we propose AFTI—a strategy that leverages facial attributes to impose explicit semantic constraints. The key contributions of this work are summarized as follows:

1. **Novel attribute-guided inversion mechanism:** We integrate a pre-trained FANet to compute attribute loss for facial template inversion, enhancing the similarity of facial attributes (e.g. eyes, nose, and mouth) between reconstructed and original images. This design imposes explicit semantic constraints, mitigating the ambiguity of template-to-image mapping. Under attribute guidance, the inverter reconstructs face images that preserve 40 facial attributes with high fidelity to the originals. To the best of our knowledge, this is the first attempt to incorporate facial attributes into facial template in-

version, which significantly improves similarity in both machine and human vision.

2. **Efficient UCNet for effective inversion:** We design UCNet, a streamlined deep network consisting of upsampling, convolution, and activation operators, to validate the effectiveness of AFTI. Despite its simplicity, images reconstructed by UCNet successfully compromise the target face recognition system. Specifically, on the LFW dataset, UCNet achieves a True Accept Rate of 0.9972 under Type-I attacks and 0.9673 under Type-II attacks at a 1% False Accept Rate. This demonstrates that a straightforward network can also achieve effective facial template inversion when guided by facial attributes.
3. **Generalizable enhancement for SOTA methods:** We integrate our proposed attribute loss into two state-of-the-art (SOTA) facial template inversion methods (Otroshi Shahreza & Marcel, 2024; Shahreza et al., 2024). Ablation experiments and comparative evaluations confirm that our attribute loss consistently boosts the performance of these baseline methods—particularly in challenging Type-II attack scenarios and across unseen recognition models—validating its strong generalization capability and universal applicability to diverse inversion frameworks.

The remainder of this paper is organized as follows. Section 2 reviews the related work on facial template inversion. In Section 3, we outline the system and the security model. Section 4 details our proposed method, followed by the experimental results presented in Section 5. Finally, the findings are summarized in Section 6.

2. Related work

The study of facial template inversion has a history that spans over two decades. In 2003, Adler (2003) introduced a method using eigenfaces to reconstruct the face image according to the subject's identity. The algorithm follows four steps: 1) build up a set of images and calculate 300 eigenfaces; 2) select an initial image from the set that has the highest matching score to the target ID by accessing the face recognition system; 3) multiply each eigenface by a small constant, add the result separately to the initial image, and choose the image with the highest score as the current one; 4) repeat the process until no further score improvement is achieved. Subsequently, Mignon and Jurie (2013) proposed another facial template inversion method, also based

on eigenfaces, where facial templates are computed using local quantized patterns. The authors trained an RBF regression model on pairs of face images and templates to create the corresponding eigenfaces, which were then used to reconstruct face images. The reconstructed images showed visual similarity to the originals and could fool some modern face recognition systems. Fredrikson et al. (2015) later proposed a method for reconstructing face images from confidence scores. This algorithm employs gradient descent to minimize a cost function tied to the face recognition model, applying denoising and sharpening to enhance image quality.

Recently, deep learning techniques have been explored for facial template inversion tasks, promising more sophisticated and effective methods for reconstructing face images from facial templates. Mai et al. (2018) proposed two types of neighborly deconvolution blocks, named NbBlock-A and NbBlock-B, to build two inverters: NbNet-A and NbNet-B. Each deconvolution block comprises a DconvOp (a cascade of deconvolution, batch-normalization and ReLU) and multiple ConvOps (a cascade of convolution, batch-normalization and ReLU). In NbBlock-A, each ConvOp only uses the output of the previous block as input, while in NbBlock-B, the ConvOps take the outputs of all preceding blocks as input, following the main idea of DenseNet (Huang et al., 2017). Both inverters are trained with the pixel and perceptual losses (Johnson et al., 2016). Shahreza et al. (2022) simplified and refined the model architecture in Mai et al. (2018) by using a combination block of one deconvolution layer, three convolution layers and a skip connection. In a subsequent work, Shahreza et al. (2024) further optimized the model by adding the skip connections within the combination blocks. They routed the output of the final combination block through four parallel convolution layers, each with different kernel sizes. The output of these layers was then merged using a sigmoid function to produce the final reconstructed image. Furthermore, they explored image reconstruction using partially leaked facial templates, demonstrating the model's ability to reconstruct faces from incomplete data (Shahreza & Marcel, 2024).

Duong et al. (2020), Truong et al. (2022) proposed a Feature-Conditional Generator (FCG) that integrates feature representations into the maps of all scales as conditional information. The loss functions are designed with the Bijective Metric to directly exploit ID distributions in the image domain. Then, the bijective distillation (similar to perceptual loss), adversary, and pixel losses are considered jointly. Dong et al. (2021) proposed a method to reconstruct high-resolution face images from facial templates using a pre-trained StyleGAN2. The authors trained a regression model to map the facial template to the latent code that can be fed into the pre-trained StyleGAN2 to generate face image. Specifically, random latent codes are generated and fed into the pre-trained StyleGAN2 to produce face images at first. Then the facial templates are extracted from these generated images using the target face recognition system. Finally, the regression model is trained using the template and latent code pairs. In their follow-up work, Dong et al. (2023) used a Genetic Algorithm (GA) generate the optimal latent code by minimizing the distance between the original template and the template extracted from the reconstructed image. Similarly, Vendrow and Vendrow (2021) used a Simulated Annealing Algorithm (SAA) to optimize the latent code. Otroshi Shahreza and Marcel (2024), Shahreza and Marcel (2023b) used StyleGAN3 to reconstruct high-resolution face images. In their approach, the authors freeze the synthesis network but train a mapping network to map the facial templates to the style vectors.

Recent advances in facial template inversion research have extended to 3D face reconstruction. Huang et al. (2023) proposed GNeRF, a method that reconstructs 3D faces from facial templates by leveraging a pre-trained EG3D model (Chan et al., 2022). Similar to prior works (Otroshi Shahreza & Marcel, 2024; Shahreza & Marcel, 2023b), their approach trains a mapping function from facial templates to the intermediate latent space of GNeRF. Furthermore, this study conducts practical presentation attacks (e.g., digital screen replay and printed photographs), systematically evaluating the vulnerability of face recognition systems to facial template inversion attacks.

All of the methods above can reconstruct face images with ID successfully preserved to some extent. The usual loss functions include pixel loss, perceptual loss, discriminate loss, and identity loss as shown in Table 1. These losses aim to achieve pixel-level accuracy and capture high-level visual understanding. However, while the reconstructed face images can bypass the target face recognition system, they often lack similarity in facial attributes to the original images. Consequently, the reconstructed images do not resemble the originals from a human perceptual perspective and exhibit unsatisfied generalization to other face recognition systems. To address this limitation, this paper introduces the Attribute-guided Facial Template Inversion (AFTI) strategy.

3. System and security model

This section defines the scenario and problems, including the system model, the attacker's knowledge, and the attacker's goals.

3.1. System model

In a face recognition system, each face image is represented by a template (a feature vector) that is expected to capture the essential characteristics of the face. Identity verification is subsequently performed by measuring the similarity between the templates of a query image and those enrolled in the system. Due to the strong correlation between the template and the identity of subject, the template must contain sensitive information about the face. This paper delves into the vulnerability of face recognition systems by demonstrating the ability to reconstruct a face image from a template.

Let us define the template (feature vector) extraction function as $E : \mathcal{X} \rightarrow \mathcal{V}$, which maps a face image $x \in \mathcal{X}$ to a template $v \in \mathcal{V}$, and denote the inversion function as $I : \mathcal{V} \rightarrow \mathcal{X}$, which aims to reverse the extraction process and reconstruct the original face image x from its template v . In our proposed scheme, the inverting function I is implemented by a generative network g_θ , referred to as the inverter, where θ represents the set of parameters in the network.

3.2. Attacker's knowledge

Here, we outline several necessary and practical assumptions to delineate the capabilities of the attacker:

1. The attacker can access the *target* face recognition system as a black box. That is, the attacker can input any face image into the target face recognition system to generate the corresponding template. In this way, the attacker can generate a large number of face image and template pairs for training. In addition, the reconstructed image could be also able to compromise the face recognition systems that are not considered in the training of inverter. We call such face recognition systems the *unseen* ones.
2. In the black-box scenario, the attacker does not know the network structure and its parameters. Thus, the attacker cannot use such knowledge to build the inverter.
3. The attacker is able to obtain the template of the target subject. This could occur through a breach in the database or due to negligence on the part of the target.

3.3. Attacker's goals

Given the template $v = E(x)$ of a face image x , the attacker in our paper aims to train a generative network g_θ with the objective function,

$$\arg \min_{\theta} (\mathcal{L}(x, x')), \quad (1)$$

where $x' = g_\theta(E(x))$. Specifically, the attacking goals can be summarized as follows.

Table 1

The loss functions used in previous methods.

Methods	Models	Target models	Loss functions	Performance
Mai et al. (2018)	NbNet	FaceNet	Pixel, Perceptual	Identity-preserving but blurry
Shahreza et al. (2024)	DSCasConvNet	Arcface-IResNet100	Pixel, Perceptual, DSSIM, Identity	
Duong et al. (2020), Truong et al. (2022)	FCG	Arcface-ResNet100	Pixel, Discrimination, Bijection, Distillation	
Dong et al. (2021)	StyleGAN2	ArcFace-ResNet100	Latent code	Identity-preserving with discrepancies in facial features
Dong et al. (2023)	GA- StyleGAN2	ArcFace-ResNet50	Latent code	
Vendrow and Vendrow (2021)	SAA- StyleGAN2	FaceNet	Latent code	
Otroshi Shahreza and Marcel (2024)	M- StyleGAN3	ArcFace-IResNet100	Pixel, Medium vector, Identity	

1. The reconstructed face image x' should be similar to the original image x of the same subject. Although previous work has typically measured image similarity using metrics such as MS-SSIM and FID, we emphasize that the most important thing is that the reconstructed image should have facial attributes similar to the original.
2. When the attacker inputs x' into the *target* face recognition system, the template $v' = E(x')$ should be similar to v to successfully bypass the verification process.
3. Furthermore, when the attacker inputs x' into an *unseen* face recognition system $\hat{E}$, the extracted template $v' = \hat{E}(x')$ is also expected to be similar to $v = \hat{E}(x)$ to ensure successful verification across multiple systems.

4. Proposed method

Facial template inversion methods aim to train a generative network g_θ that reconstructs face images from facial templates. As deep learning-based approaches, their critical components include network architecture, loss function, and training data. Prior works have designed sophisticated network architectures trained on large-scale, well-established face image datasets. While existing methods can generate reconstructed face images that bypass target face recognition systems, these images often lack similarity in key facial attributes to the originals. Consequently, they do not resemble the original faces from a human perceptual perspective and exhibit poor generalization to *unseen* face recognition systems.

To address this limitation, this paper proposes a facial attribute loss to guide inverters in generating face images that better preserve critical facial attributes (e.g., eyes, nose, and mouth) relative to the original images. To validate the effectiveness of our approach, we develop UCNet—a streamlined yet powerful network architecture that employs basic up-sampling, convolution, and activation operations. Furthermore, we successfully enhance two state-of-the-art frameworks—M-StyleGAN3 and DSCasConvNet (just discussed in experiments) by incorporating our attribute guidance. These improvements underscore that attribute guidance, which is independent of model architecture, is the key driver of enhanced inversion performance.

4.1. UCNet for facial template inversion

Numerous generative networks have been employed for facial template inversion, including NbNet (Mai et al., 2018), StyleGAN (Dong et al., 2021, 2023; Otroshi Shahreza & Marcel, 2024; Shahreza & Marcel, 2023b; Vendrow & Vendrow, 2021), and others. To underscore the effec-

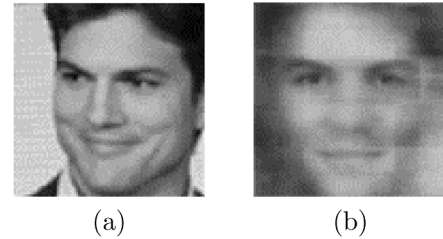


Fig. 4. Checkerboard pattern of artifacts caused by uneven overlap in deconvolution: (a) An original face image, and (b) The corresponding face image reconstructed by deconvolution.

tiveness of the proposed attribute loss, this paper introduces UCNet—a streamlined deep network whose core building blocks are UCBlocks.

Given the objective of reconstructing face images from facial templates, deconvolution operations may seem like a natural choice for our network architecture. However, prior work by Odena et al. (2016) has highlighted that deconvolution often leads to checkerboard artifacts due to uneven overlap, which degrades the quality of reconstructed images (see Fig. 4). To mitigate this issue, we propose UCBlock and UCBlock-out, which serve as the core building blocks for constructing UCNet. As illustrated in Fig. 5, these proposed blocks comprise an upsampling layer, a Conv2d layer, a BatchNorm2d layer, and a Tanh or Sigmoid activation layer (depending on the block type).

In our method, we first assume a 512-dimensional facial template for simplification—consistent with the output dimensions of many widely used face recognition models (e.g., ArcFace Dong et al., 2023 and AdaFace Kim et al., 2022). We then reshape the template into a tensor of shape $512 \times 1 \times 1$, where 512 denotes the number of channels. Both UCBlock and UCBlock-out integrate an upsampling operator that uniformly doubles the width and height of the tensor. The parameters of the Conv2d convolution operator are detailed in Table 2. Additionally, our network employs standard operations (e.g., BatchNorm2d, Tanh, and Sigmoid) that do not modify the tensor dimensions. This carefully designed structure enables UCNet to invert a 512-dimensional template into a 128×128 color face image which is sufficiently large to clearly depict facial details. Notably, by adjusting the number of UCBlocks, our network can be flexibly adapted to generate face images of any desired resolution.

4.2. Loss functions of UCNet

Our scheme incorporates four types of loss to guarantee the quality of the reconstructed image, including pixel loss $\mathcal{L}_{pix}$, discrimination

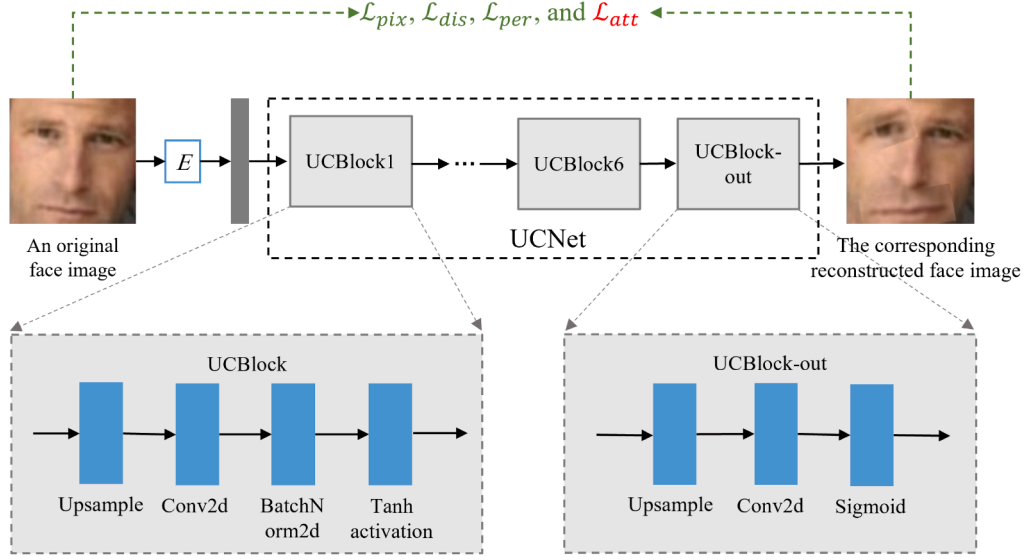


Fig. 5. The structure of UCNet.

Table 2
The details of UCNet.

UCBlocks	The parameters of Conv2D (kernel size, kernel numbers, stride, padding)	Output size ($c \times w \times h$)
UCBlock1	$[3 \times 3 \times 512]$, 4096, 1, 1	$4096 \times 2 \times 2$
UCBlock2	$[3 \times 3 \times 4096]$, 2048, 1, 1	$2048 \times 4 \times 4$
UCBlock3	$[3 \times 3 \times 2048]$, 1024, 1, 1	$1024 \times 8 \times 8$
UCBlock4	$[3 \times 3 \times 1024]$, 512, 1, 1	$512 \times 16 \times 16$
UCBlock5	$[3 \times 3 \times 512]$, 256, 1, 1	$256 \times 32 \times 32$
UCBlock6	$[3 \times 3 \times 256]$, 128, 1, 1	$128 \times 64 \times 64$
UCBlock-out	$[3 \times 3 \times 128]$, 3, 1, 1	$3 \times 128 \times 128$

Table 3
The details for discriminator.

Layer name	Kernel size, kernel numbers, stride, padding	Output size ($c \times w \times h$)
Conv2d-1	$[5 \times 5 \times 3]$, 64, 2, 2	$64 \times 64 \times 64$
Conv2d-2	$[5 \times 5 \times 64]$, 128, 2, 2	$128 \times 32 \times 32$
Conv2d-end	$[32 \times 32 \times 128]$, 1, 1, 0	$1 \times 1 \times 1$

loss $\mathcal{L}_{dis}$, perceptual loss $\mathcal{L}_{per}$, and attribute loss $\mathcal{L}_{att}$. The total loss is calculated as follows.

$$\mathcal{L} = \lambda_{pix} \mathcal{L}_{pix} + \lambda_{dis} \mathcal{L}_{dis} + \lambda_{per} \mathcal{L}_{per} + \lambda_{att} \mathcal{L}_{att}, \quad (2)$$

where λ_{pix} , λ_{per} , λ_{dis} , and λ_{att} represent the weights of each loss component.

4.2.1. Pixel loss

The pixel loss aims to ensure that the reconstructed image closely resembles the original image at the pixel level. It is computed directly in the spatial domain and is a common loss function in many image generation tasks. Formulaically, we use the Mean Square Error (MSE) as our pixel loss metric:

$$\mathcal{L}_{pix} = \frac{\sum_{i=1}^n (p_i - p'_i)^2}{n}, \quad (3)$$

where p_i and p'_i denote the pixel values in the original x and reconstructed images x' , respectively, and n represents the total number of pixels in the images.

4.2.2. Discrimination loss

The discrimination loss is designed to ensure that the reconstructed and original images are indistinguishable from each other by the ad-

versarially trained discriminator. To achieve this, we have developed a binary classification network $D(\cdot)$, as illustrated in Fig. 6 and detailed in Table 3. This network is trained adversarially with UCNet. During the training process, the network $D(\cdot)$ is updated with the following loss function:

$$\begin{aligned} \mathcal{L}_{disD} = & -\frac{1}{n} \sum_{i=1}^n (y_i \log D(x) + (1 - y_i) \log(1 - D(x)) \\ & + y_i \log(D(x')) + (1 - y_i) \log(1 - D(x'))) \\ = & -\frac{1}{n} \sum_{i=1}^n (\log D(x) + \log(1 - D(x'))), \end{aligned} \quad (4)$$

where $y_i \in \{0, 1\}$ with the value 1 denoting the original face image and the value 0 denoting the reconstructed image, and n denotes the batch size. In the context of updating UCNet, the discrimination loss is defined as follows:

$$\mathcal{L}_{dis} = -\frac{1}{n} \sum_{i=1}^n (\log D(x')). \quad (5)$$

4.2.3. Perceptual loss

The computer vision community has revealed that internal feature maps from convolution networks, which are trained on advanced image classification tasks, are remarkably effective perceptual representations. Following the recommendation of Zhang et al. (2018), we use feature maps from 5 layers within the VGG-16 architecture to calculate the perceptual loss, as depicted in Fig. 7. For the l th layer, the feature maps of the original and reconstructed images are denoted as $m_{hw}^l, m'_{hw}^l \in \mathbb{R}^{H_l \times W_l \times C_l}$, where H_l and W_l represent the spatial dimensions and C_l signifies the channel dimension. m_{hw}^l and m'_{hw}^l will be unit-normalized at first, and then the perceptual loss can be calculated as follows:

$$\mathcal{L}_{per} = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w^l \odot (m_{hw}^l - m'_{hw}^l)\|_2^2, \quad (6)$$

where $\odot$ denotes dot product, and $w^l \in \mathbb{R}^{C_l}$ is the fixed parameter that is determined during the training of VGG-16 (Zhang et al., 2018).

4.2.4. Attribute loss

Humans typically recognize individuals by observing their facial characteristics, such as eyes, nose, mouth, and other distinctive features. For example, when verifying the identity of a face in a photograph, we

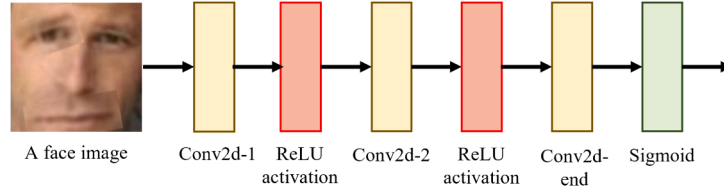


Fig. 6. The architecture of discriminator.



Fig. 7. Architecture of VGG-16: Feature maps are extracted from 5 layers (ReLU1-2, ReLU2-2, ReLU3-3, ReLU4-3, and ReLU5-3).

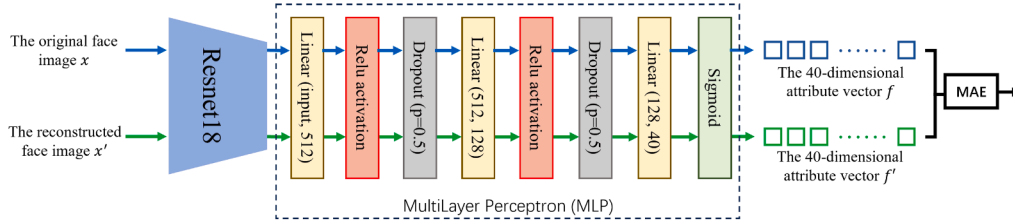


Fig. 8. Architecture of FANet, constructed with ResNet18 and an MLP. Our attribute loss is defined as the MAE between feature vectors extracted from FANet.

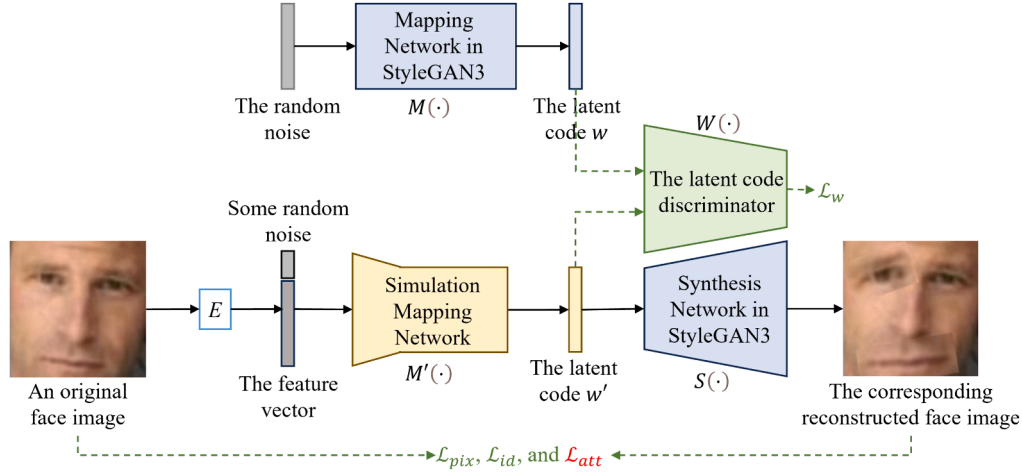


Fig. 9. The structure of attribute-guided M-StyleGAN3.

often compare these specific facial traits for consistency. Consequently, we designed the attribute loss to ensure that the reconstructed image retains facial attributes similar to those of the original. Unlike identity loss (i.e., template loss) that primarily measures high-level feature similarity using facial templates, our attribute loss enhances the similarity of discriminative facial attributes (e.g., eyes, nose, and mouth) that are naturally salient to human perception. This attribute-guided constraint introduces explicit semantic supervision to the inversion process, effectively mitigating the ambiguity inherent in template-to-image mapping.

To implement this attribute loss, we utilize a pre-trained facial attribute analysis model,² denoted as FANet, to compute the attribute loss. Adapted from ResNet18 (He et al., 2016) and fine-tuned on the CelebA dataset (Liu et al., 2015) (see Fig. 8), FANet consists of a ResNet18 backbone integrated with a MultiLayer Perceptron (MLP) (Riedmiller & Lerner, 2014). Specifically, this combined architecture is trained on CelebA to output a 40-dimensional facial attribute vector, which cap-

tures discriminative semantic features (e.g. eye shape, nose structure, mouth characteristics). The attribute loss is then calculated using Mean Absolute Error (MAE), defined as follows:

$$\mathcal{L}_{att} = \frac{\sum_{i=1}^{|f|} |f(i) - f'(i)|}{|f|}, \quad (7)$$

where f and f' denote the facial attribute vectors extracted from the original and generated images, respectively, and $f(i)$ and $f'(i)$ represent their i th elements.

4.3. Attribute-guided M-StyleGAN3 for facial template inversion

Recent StyleGAN-based methods (Dong et al., 2021, 2023; Shahreza & Marcel, 2023b; Vendrow & Vendrow, 2021) have demonstrated strong performance in facial template inversion. However, they exhibit low True Accept Rates (TARs) under Type-II attacks and unsatisfied generalization capability when applied to unseen recognition models. To overcome these challenges, we propose AFTI, which leverages attribute loss

² <https://github.com/Hawaii0821/FaceAttr-Analysis>.

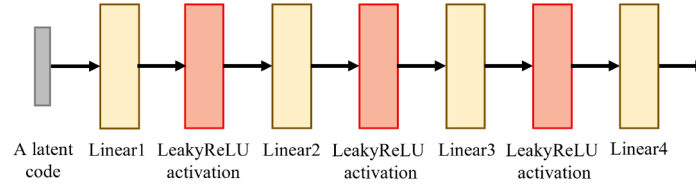


Fig. 10. The structure of the latent code discriminator.

to guide the inversion process using M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) (currently the state-of-the-art method in this field).

As depicted in Fig. 9, the attribute-guided M-StyleGAN3 comprises a simulation mapping network and the pre-trained synthesis network from StyleGAN3. The simulation mapping network consists of two linear layers: the first takes a 512-dimensional facial template (feature vector) and an 8-dimensional random noise vector as inputs, outputting a 512-dimensional intermediate feature; the second processes this intermediate feature to generate a 512-dimensional latent code z' . To align the distribution of z' with that of the latent code z from StyleGAN3's original mapping network, we introduce a latent code discriminator (see Fig. 10).

To train a facial template inversion model, we treat the synthesis network from StyleGAN3 (Karras et al., 2021) as a fixed, pre-trained face generation network, and only train the simulation mapping network with four loss functions: the latent code discrimination loss $\mathcal{L}_w$, pixel loss $\mathcal{L}_{pix}$, identity loss $\mathcal{L}_{id}$, and attribute loss $\mathcal{L}_{att}$.

The pixel loss $\mathcal{L}_{pix}$ and attribute loss $\mathcal{L}_{att}$ adopt the same formulation as described in Section 4.2. The latent code discrimination loss is similar to described in Section 4.2 but differs in architecture and discriminated targets. The identity loss $\mathcal{L}_{id}$ is computed by extracting ElasticFace features (Boutros et al., 2022) from the generated image and the original one, and then calculating the MSE between them.

5. Experimental results

To validate the effectiveness of our proposed approach, we conducted comprehensive evaluations on the similarity between reconstructed and original images using conventional evaluation metrics. Additionally, we fed the reconstructed images into both *target* and *unseen* face recognition systems to assess their identity-preserving performance.

5.1. Experiment configurations and datasets

All experiments were conducted using PyTorch on a workstation equipped with two NVIDIA RTX 3090 GPUs. We selected FaceNet (Schroff et al., 2015), ArcFace-IRNet100 (Deng et al., 2019), and SwinFace (Qin et al., 2024) as the target face recognition models. The resolution of reconstructed face images was fixed at 128×128 pixels. For model training, we adopted a batch size of $b = 32$ and an initial learning rate of $\gamma = 0.01$; the loss weighting parameters were set as $\lambda_{pix} = \lambda_{per} = \lambda_{dis} = 1$, while λ_{att} was set to 10 to amplify the impact of facial attribute supervision.

In our experiments, we utilized four representative datasets: FFHQ (Karras et al., 2021), DCGAN-VGGFace (Mai et al., 2018), LFW (Learned-Miller et al., 2016), and AgeDB (Moschoglou et al., 2017). FFHQ (Flickr-Faces-High-Quality) comprises 70,000 high-quality face images without identity annotations. DCGAN-VGGFace consists of synthetic images generated by DCGAN (Radford, 2015), which was pre-trained on the VGGFace dataset (Parkhi et al., 2015) by Mai et al. (2018). VGGFace contains approximately 2.6 million face images from 2622 unique individuals, and DCGAN-VGGFace (Mai et al., 2018) serves as an effective data augmentation strategy for VGGFace. LFW (Labeled Faces in the Wild) consists of 13,233 face images from 5749 distinct individuals, exhibiting diversity across pose, facial expression, ethnicity, illumination, focus, and image resolution. AgeDB (Moschoglou et al., 2017) contains

16,488 images of 568 famous individuals, characterized by substantial variations in age, illumination, and pose, showing greater challenges for facial template inversion compared to LFW.

5.2. Evaluation metrics

Three metrics are employed for similarity evaluation.

5.2.1. FAMSE

We propose the Facial Attribute Mean Squared Error (FAMSE) metric to quantify the similarity of facial attributes between images, with a specific focus on the eye, nose, and mouth regions. These regions can be localized using facial landmarks (Zhang et al., 2016a), and FAMSE is then computed as follows:

$$\begin{aligned} \text{FAMSE}(x, x') = & \frac{1}{3} \sum_{i=1}^{n_{\text{eyes}}} \frac{(p_i^{\text{eyes}} - p_i^{\text{eyes}'})^2}{n_{\text{eyes}}} \\ & + \frac{1}{3} \sum_{i=1}^{n_{\text{nose}}} \frac{(p_i^{\text{nose}} - p_i^{\text{nose}'})^2}{n_{\text{nose}}} \\ & + \frac{1}{3} \sum_{i=1}^{n_{\text{mouth}}} \frac{(p_i^{\text{mouth}} - p_i^{\text{mouth}'})^2}{n_{\text{mouth}}} \end{aligned} \quad (8)$$

FAMSE values range from 0 to 1 with lower values indicating higher similarity between the images.

5.2.2. Deep-feature similarity

Deep-Feature Similarity (DFS) quantifies how well reconstructed images preserve the identity information of original images at the feature level. It measures the similarity of deep features extracted from seven representative face recognition models: FaceNet (Schroff et al., 2015), ArcFace (Deng et al., 2019), SwinFace (Qin et al., 2024), AdaFace (Kim et al., 2022), MagFace (Meng et al., 2021), SphereFace (Liu et al., 2017), and CurricularFace (Huang et al., 2020). For each model E_k (where $k = 1, 2, \dots, 7$), we extract the feature templates $v_k = E_k(x)$ and $v'_k = E_k(x')$ corresponding to the original image x and its reconstructed version x' , respectively. The cosine similarity between the original and reconstructed feature templates for each model is then computed as:

$$\text{CS}_k(x, x') = \frac{v_k \cdot v'_k}{\|v_k\| \times \|v'_k\|}, \quad (9)$$

where $\cdot$ represents the dot product, and $\|\cdot\|$ represents the Euclidean norm. Then overall DFS is obtained by averaging the cosine similarities across all seven models:

$$\text{DFS}(x, x') = \frac{1}{7} \sum_{k=1}^7 \text{CS}_k(x, x'). \quad (10)$$

DFS values range from -1 to 1, where higher values indicate superior preservation of the original identity information in reconstructed images. This multi-model metric approach enables a robust and comprehensive assessment of identity consistency, as it aggregates discriminative evidence from diverse template spaces across state-of-the-art face recognition systems.

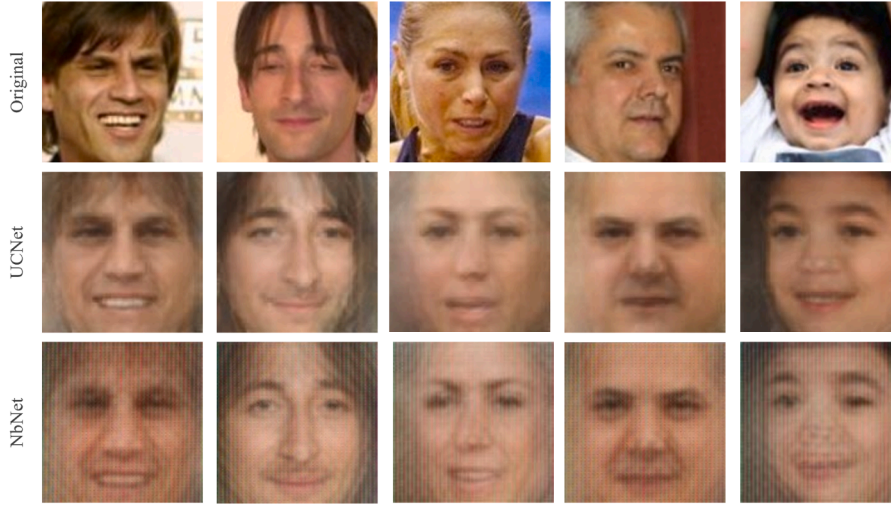


Fig. 11. Visual examples of reconstructed face images, tested on LFW with FaceNet as the target model.

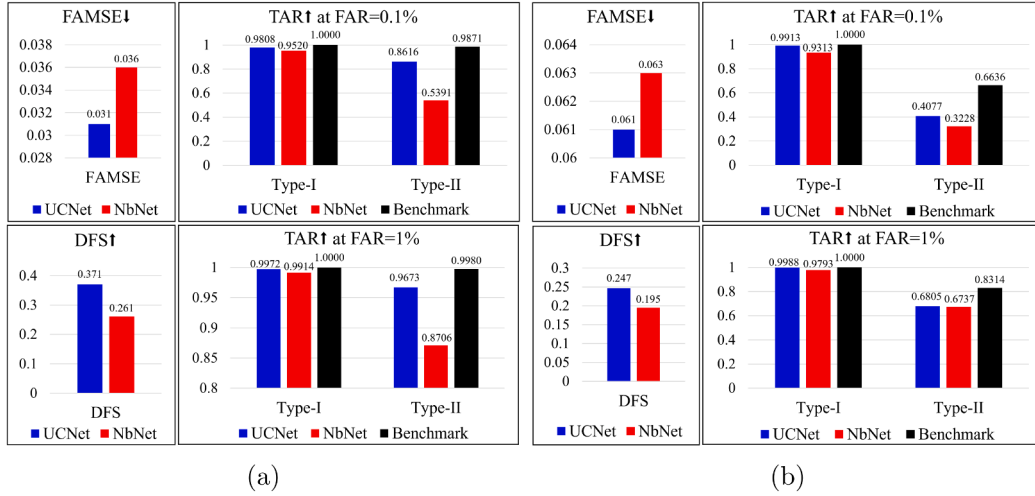


Fig. 12. Performance comparison of UCNet and NbNet (Mai et al., 2018) on (a) LFW and (b) AgeDB, using FaceNet as the target model.

5.2.3. Verification precision by the face recognition system

The reconstructed image is expected to pass through the face recognition system successfully. The BLUFR protocol (Liao et al., 2014) is used to evaluate the precision of verification for fair comparison. Let $\mathcal{G}$ denote the set of gallery images that are enrolled in the face recognition system. We further define two probe image sets: $\mathcal{P}_G$ and $\mathcal{P}_N$. $\mathcal{P}_G$ consists of images of subjects that are present in the gallery set $\mathcal{G}$ but with different images, while $\mathcal{P}_N$ includes images of subjects that are not in $\mathcal{G}$.

The verification precision is then evaluated using the False Accept Rate (FAR) and True Accept Rate (TAR), which are defined as follows:

$$\begin{cases} \text{FAR}(\tau) = \frac{|\{\forall p \in \mathcal{P}_N | \max_{g \in \mathcal{G}} \text{CS}(g, p) > \tau\}|}{|\mathcal{P}_N|}, \\ \text{TAR}(\tau) = \frac{|\{\forall p \in \mathcal{P}_G | \max_{g \in \mathcal{G}} \text{CS}(g, p) > \tau\}|}{|\mathcal{P}_G|}. \end{cases} \quad (11)$$

In line with previous work (Mai et al., 2018) and (Dong et al., 2023), we calculate the verification precision in two cases:

- Type-I attack involves matching the reconstructed images with the face images from which the templates are produced. We also report the benchmark with the original face images (instead of the recon-

structed images) as the query. In Type-I attack, the benchmark obviously remains 100%.

- Type-II attack refer to matching the reconstructed image with a face image of the same subject, but it was not the one used to produce the template. Again, we report the benchmark using original face images as the query for comparison.

5.3. Results with FaceNet as the target model

Here, we present experimental results with FaceNet as the target model, comparing our proposed UCNet against the state-of-the-art (SOTA) method NbNet (Mai et al., 2018). For fair comparison, our UCNet is trained under the same experimental settings as NbNet, and the results of NbNet are directly copied from its original paper (Mai et al., 2018).

Target model. FaceNet is a classical facial template generation method that served as the target model in NbNet (Mai et al., 2018). It outputs 128-dimensional facial templates; consequently, we adapt the input dimension of UCNet to 128 by modifying the kernel size in UCBLOCK1 to $[3 \times 3 \times 128]$.

Training set. Consistent with NbNet (Mai et al., 2018), synthetic images generated by a pre-trained DCGAN are used to train UCNet. Prior to training, these generated images are aligned using five facial

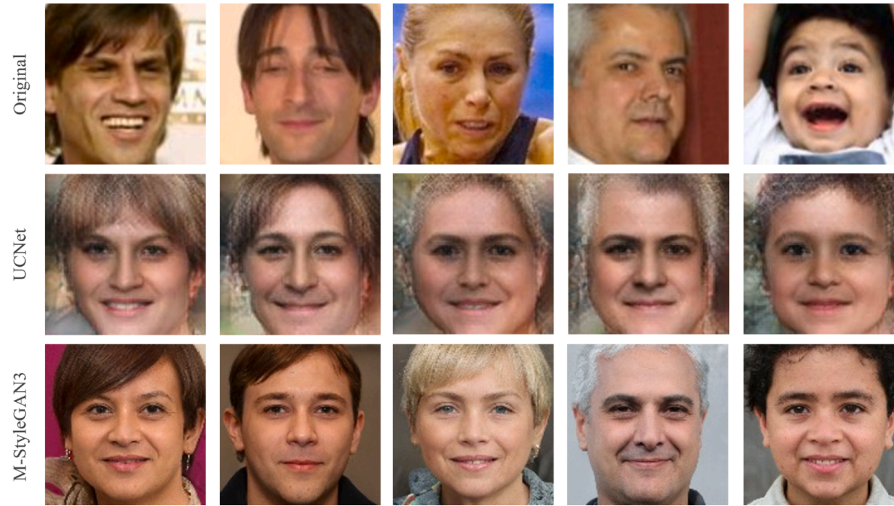


Fig. 13. Visual examples of reconstructed face images, tested on LFW with ArcFace-IResNet100 as the target model.

landmarks detected by MTCNN³ (Zhang et al., 2016b), and then cropped to 160×160 pixels—following the same preprocessing pipeline as NbNet (Mai et al., 2018). Since UCNet is designed to generate 128×128 images, the DCGAN-generated images are further resized to 128×128 pixels for training.

Testing set. As in NbNet (Mai et al., 2018), the LFW dataset is adopted as the primary testing set. Additionally, we introduce a second in-the-wild dataset (AgeDB) to evaluate generalization. Consistent with the training preprocessing, images from both LFW and AgeDB are aligned, cropped, and resized to 128×128 pixels before being fed into FaceNet for feature vector extraction during UCNet’s testing phase.

Results. We first present qualitative results of reconstructed faces from the LFW dataset. As illustrated in Fig. 11, faces generated by UCNet exhibit higher clarity compared to those produced by NbNet (Mai et al., 2018). Quantitative results for FAMSE, DFS, and verification precision are reported in Fig. 12, where NbNet’s verification precision is directly copied from its original paper (Mai et al., 2018). Benchmark values are computed using original images as queries. As shown in Fig. 12, UCNet achieves superior FAMSE and DFS scores—quantitatively confirming that its generated faces are more similar to original images than those of NbNet (Mai et al., 2018). Furthermore, UCNet outperforms NbNet in verification precision when evaluated on the target face recognition system, with particularly notable superiority under Type-II attack scenarios.

5.4. Results with ArcFace-IResNet100 as the target model

Here, we present test results with ArcFace-IResNet100 as the target model, comparing our proposed UCNet against a StyleGAN3-based facial template inversion method (Otroshi Shahreza & Marcel, 2024)—referred to as M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) for simplicity. This method trains a novel mapping network to map facial templates to style vectors while keeping the synthesis network unchanged. For fair comparison, UCNet is trained under the same experimental settings as M-StyleGAN3.

Target model. ArcFace is a highly influential deep learning-based face recognition method in the computer vision field. M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) adopted ArcFace-IResNet100 as its target model—an advanced deep face recognition architecture with IResNet100 as the backbone, trained with the ArcFace loss function. It outputs 512-dimensional facial templates.

Training set. Consistent with M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024), the FFHQ dataset is used to train UCNet. Prior to training, FFHQ face images are cropped based on facial landmarks and resized to 128×128 pixels. These preprocessed images are then fed into ArcFace-IResNet100 to generate 512-dimensional facial templates. Finally, pairs of resized images and their corresponding templates are used as training data for UCNet.

Testing set. As in M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024), the LFW dataset is adopted for UCNet’s testing. Additionally, we introduce the AgeDB dataset to further enrich the experimental evaluation. Consistent with the training set preprocessing pipeline, images from both LFW and AgeDB are aligned, cropped and resized to 128×128 pixels.

Results. We first present qualitative examples of reconstructed face images. As shown in Fig. 13, M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) generates face images with high resolution and visual quality. However, UCNet’s generated faces exhibit higher similarity to the original images—quantitatively validated by the FAMSE and DFS metrics (see Fig. 14). Furthermore, UCNet outperforms M-StyleGAN3 in verification precision when evaluated on the ArcFace-IResNet100 system, with particularly notable superiority under Type-II attack scenarios (see Fig. 14). Note that the TARs of M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) are

obtained using the authors’ pre-trained model,⁴ and all TAR evaluations are performed on the same testing sets following the BLUFR protocol (Liao et al., 2014).

5.5. Results with SwinFace as the target model

We further evaluate the performance when adopting SwinFace as the target model. As established in Sections 5.3 and 5.4, M-StyleGAN3 already outperforms NbNet in facial template inversion tasks. Therefore, this subsection focuses on comparing our proposed UCNet and attribute-guided M-StyleGAN3 against M-StyleGAN3.

Target model. SwinFace is a transformer-based multi-task framework designed for face recognition, expression recognition, age estimation, and attribute estimation, outputting 512-dimensional facial templates. For consistency, UCNet, M-StyleGAN3, and attribute-guided M-StyleGAN3 are trained under the same experimental settings as those adopted for ArcFace-IResNet100 (Section 5.4), with the only differences being the target facial templates and the identity loss function.

³ https://github.com/YYuanAnyVision/mxnet_mtcnn_face_detection.

⁴ https://github.com/bob/bob.paper.neurips2023_face_ti/-/tree/master/checkpoints?ref_type=heads.

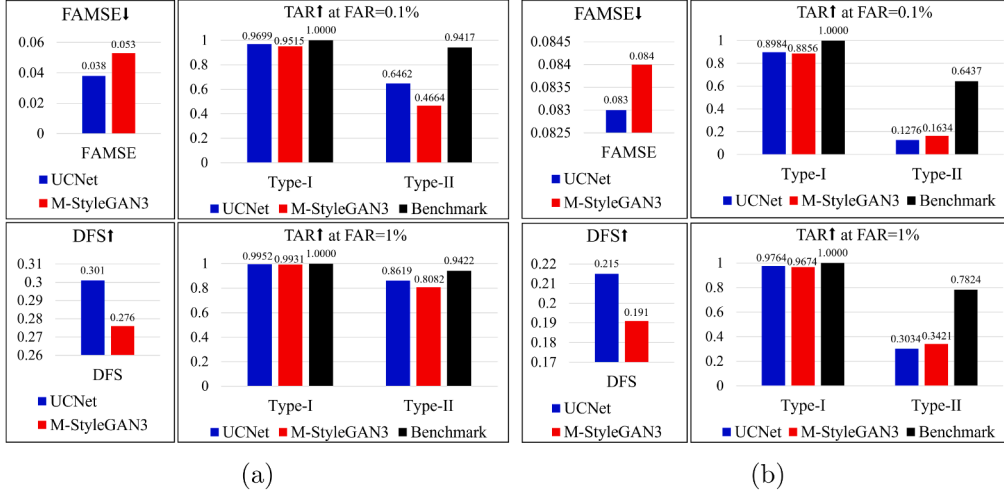


Fig. 14. Performance comparison of UCNet and M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) on (a) LFW and (b) AgeDB, using ArcFace-IResNet100 as the target model.



Fig. 15. Visual examples of reconstructed face images, tested on LFW with SwinFace as the target model.

Training set. Consistent with the M-StyleGAN3 comparison setup, the training set follows the identical configuration as described in Section 5.4.

Testing set. Similarly, the testing set adheres to the same configuration as detailed in Section 5.4.

Results. Qualitative examples of reconstructed face images from the LFW dataset are provided in Fig. 15. As illustrated, face images generated by attribute-guided inverters exhibit higher similarity to those produced by M-StyleGAN3. Quantitative results for FAMSE, DFS, and verification precision are reported in Fig. 16. The results indicate that the attribute-guided facial template inverters achieve superior performance across all metrics. This not only quantitatively validates that the images generated by these attribute-guided models maintain higher similarity to the original images but also underscores the value of integrating attribute guidance in strengthening the accuracy of facial template inversion.

5.6. Ablation experiments on $\mathcal{L}_{att}$

To further validate the effectiveness of our proposed AFTI, we conducted the following ablation experiments on the LFW dataset.

1. **UCNet-FaceNet with/without attribute guidance:** We evaluate the performance of UCNet trained with/without the attribute loss $\mathcal{L}_{att}$ (i.e., with or without attribute guidance), using FaceNet as the target model.
2. **UCNet-ArcFace with/without attribute guidance:** Similarly, we assess the performance of UCNet trained with/without $\mathcal{L}_{att}$, where ArcFace-IResNet100 serves as the target model.
3. **UCNet-SwinFace with/without attribute guidance:** Consistent with the above settings, we test the performance of UCNet trained with/without $\mathcal{L}_{att}$, adopting SwinFace as the target model.

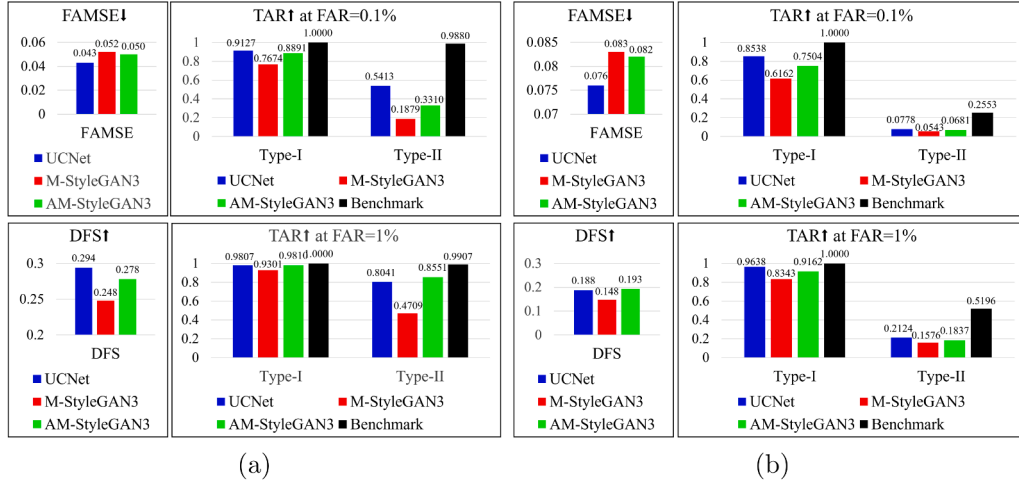


Fig. 16. Performance comparison of UCNet, M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024), and attribute-guided M-StyleGAN3 (AM-StyleGAN3) on (a) LFW and (b) AgeDB, using SwinFace as the target model.

Table 4

The ablation study of $\mathcal{L}_{att}$ on UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace, using LFW as the testing dataset.

Models	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
UCNet-FaceNet	✓	0.9808	0.8616	0.9972	0.9673
	✗	0.9763	0.8154	0.9943	0.9546
UCNet-ArcFace	✓	0.9699	0.6462	0.9952	0.8619
	✗	0.3404	0.0731	0.6378	0.3340
UCNet-SwinFace	✓	0.8549	0.3928	0.9598	0.6907
	✗	0.3469	0.1543	0.6140	0.3918
M-StyleGAN3-ArcFace	✓	0.8958	0.4987	0.9992	0.9322
	✗	0.9515	0.4664	0.9931	0.8082
M-StyleGAN3-SwinFace	✓	0.8891	0.3310	0.9810	0.6995
	✗	0.7674	0.1879	0.9301	0.4709

Table 5

The FAMSE of UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace trained with/without $\mathcal{L}_{att}$, using LFW as the testing dataset.

Models	$\mathcal{L}_{att}$	FAMSE
UCNet-FaceNet	✓	0.031
	✗	0.045
UCNet-ArcFace	✓	0.038
	✗	0.052
UCNet-SwinFace	✓	0.043
	✗	0.043
M-StyleGAN3-ArcFace	✓	0.037
	✗	0.053
M-StyleGAN3-SwinFace	✓	0.050
	✗	0.052

- M-StyleGAN3-ArcFace with/without attribute guidance:** As a strong baseline in facial template inversion, M-StyleGAN3 generates face images with higher resolution and quality owing to the inherent capabilities of StyleGAN3. To verify the generalizability of our attribute guidance, we integrate $\mathcal{L}_{att}$ into M-StyleGAN3, with ArcFace-IResNet100 as the target model.
- M-StyleGAN3-SwinFace with/without attribute guidance:** Consistent with the setup of M-StyleGAN3-ArcFace, we test the performance of M-StyleGAN3 trained with/without $\mathcal{L}_{att}$, adopting SwinFace as the target model.

As shown in Tables 4 and 5, the attribute guidance consistently improves the inversion performance across all tested models. This confirms that our AFTI can effectively enhance the model's ability to recover discriminative facial features, regardless of the base network architecture or target face recognition model.

5.7. Results on unseen models

Here, we investigate whether face images reconstructed against a given target face recognition model can generalize to and bypass *unseen* face recognition systems. For this evaluation, four state-of-the-art *un-*

seen face recognition models are employed: AdaFace⁵ (Kim et al., 2022), MagFace⁶ (Meng et al., 2021), SphereFace⁷ (Liu et al., 2017), and CurricularFace⁸ (Huang et al., 2020). These models are designated as *unseen* because they were not involved in guiding the training of the inversion models, ensuring a rigorous assessment of cross-model generalization.

We evaluate the generalization capability of five inversion models: UCNet-FaceNet, UCNet-ArcFace, M-StyleGAN3-ArcFace, UCNet-SwinFace, and M-StyleGAN3-SwinFace. As presented in Tables 6 to 9, the inversion models trained with the attribute loss $\mathcal{L}_{att}$ consistently achieve higher TARs compared to their counterparts trained without $\mathcal{L}_{att}$. This result confirms that attribute guidance effectively enhances the cross-model generalizability of reconstructed faces, enabling them to reliably bypass diverse unseen face recognition systems.

⁵ <https://github.com/mk-minchul/AdaFace>.

⁶ <https://github.com/IrvingMeng/MagFace>.

⁷ <https://github.com/wyliu/sphereface>.

⁸ <https://github.com/HuangYG123/CurricularFace>.

Table 6

TARs of UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace trained with/without $\mathcal{L}_{att}$, tested on LFW with AdaFace as the *unseen* model.

Models	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
UCNet-FaceNet	✓	0.7025	0.4491	0.8775	0.7392
	✗	0.6705	0.4344	0.8644	0.7364
UCNet-ArcFace	✓	0.8549	0.3928	0.9598	0.6907
	✗	0.3469	0.1543	0.6140	0.3918
UCNet-SwinFace	✓	0.4483	0.1810	0.7014	0.4586
	✗	0.4425	0.1169	0.6980	0.3274
M-StyleGAN3-ArcFace	✓	0.5566	0.2800	0.7762	0.5940
	✗	0.3526	0.0562	0.5840	0.1910
M-StyleGAN3-SwinFace	✓	0.3077	0.0468	0.5366	0.1501
	✗	0.2262	0.0318	0.4451	0.1114

Table 7

TARs of UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace trained with/without $\mathcal{L}_{att}$, tested on LFW with MagFace as the *unseen* model.

Models	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
UCNet-FaceNet	✓	0.7761	0.6447	0.9120	0.8461
	✗	0.7680	0.6115	0.9091	0.8331
UCNet-ArcFace	✓	0.8958	0.4987	0.9752	0.7890
	✗	0.7656	0.3337	0.9248	0.6418
UCNet-SwinFace	✓	0.6093	0.3326	0.8317	0.6593
	✗	0.6088	0.2176	0.8297	0.4750
M-StyleGAN3-ArcFace	✓	0.9565	0.7567	0.9929	0.9373
	✗	0.8466	0.3677	0.9675	0.7704
M-StyleGAN3-SwinFace	✓	0.6348	0.2285	0.8551	0.6137
	✗	0.5176	0.1530	0.7659	0.4367

Table 8

TARs of UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace trained with/without $\mathcal{L}_{att}$, tested on LFW with SphereFace as the *unseen* model.

Models	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
UCNet-FaceNet	✓	0.5070	0.2792	0.7030	0.5102
	✗	0.4876	0.2578	0.6853	0.4867
UCNet-ArcFace	✓	0.2679	0.0552	0.5613	0.3656
	✗	0.1458	0.0173	0.3695	0.1285
UCNet-SwinFace	✓	0.5406	0.2533	0.7831	0.5672
	✗	0.5393	0.1552	0.7811	0.3887
M-StyleGAN3-ArcFace	✓	0.6205	0.3430	0.8263	0.7039
	✗	0.4175	0.0837	0.6517	0.2793
M-StyleGAN3-SwinFace	✓	0.3296	0.0596	0.5560	0.1762
	✗	0.2412	0.0418	0.4656	0.1272

Table 9

TARs of UCNet-FaceNet, UCNet-ArcFace, UCNet-SwinFace, M-StyleGAN3-ArcFace, and M-StyleGAN3-SwinFace trained with/without $\mathcal{L}_{att}$, tested on LFW with CurricularFace as the *unseen* model.

Models	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
UCNet-FaceNet	✓	0.5362	0.2948	0.6855	0.4594
	✗	0.5243	0.2905	0.6757	0.4568
UCNet-ArcFace	✓	0.3197	0.1961	0.6144	0.5900
	✗	0.1614	0.0338	0.3856	0.2289
UCNet-SwinFace	✓	0.6028	0.3583	0.8288	0.6610
	✗	0.6025	0.2345	0.8201	0.5023
M-StyleGAN3-ArcFace	✓	0.9190	0.6859	0.9807	0.9044
	✗	0.7774	0.3752	0.9234	0.7534
M-StyleGAN3-SwinFace	✓	0.5942	0.2092	0.8016	0.4967
	✗	0.4687	0.1138	0.7017	0.3275

5.8. Tests with DSCasConvNet

To further validate the generalizability of our proposed AFTI strategy across diverse network architectures, we extended our evaluation to DSCasConvNet (Shahreza et al., 2024)—a state-of-the-art custom-built network (in contrast to pre-trained generative models like M-StyleGAN3). Specifically, we retrained DSCasConvNet with our attribute loss $\mathcal{L}_{att}$ to assess the effectiveness of AFTI, using ArcFace-IRNet100 as the target face recognition model. As presented in Table 10 and Fig. 17, the results demonstrate that our AFTI strategy consistently enhances the inversion performance of DSCasConvNet, further confirming its strong generalizability across different base network designs.

5.9. Image similarity in human visual perception

This section evaluates whether the attribute loss $\mathcal{L}_{att}$ can enhance the perceived similarity of reconstructed faces in human visual perception. A questionnaire consisting of 40 single-choice questions was designed, as illustrated in Fig. 18(a). For each question, participants were presented with an original face image (from the LFW dataset) and tasked with selecting the image that was more visually similar to the original from two reconstructed candidates: one generated by UCNet trained with the attribute loss $\mathcal{L}_{att}$, and the other by UCNet trained without $\mathcal{L}_{att}$. To screen out inattentive responses, two sentinel questions with intuitively obvious answers were additionally included in the questionnaire.

The questionnaire was distributed via QR code on WeChat. Participants were notified that their data would be used solely for academic research, and no identifiable personal information was collected. Two sentinel questions with clear ground-truth answers were embedded for validity filtering. The display positions of the two reconstruction alternatives were fully randomized per respondent. In total, we received 280 anonymous responses. After discarding invalid submissions with inconsistent sentinel-question answers, 220 valid responses remained. Since survey content is randomly regenerated for each submission, multiple submissions from the same participant are treated as independent and valid observations in the analysis. The results are summarized in Fig. 18(b), which demonstrate that the reconstructed images generated by UCNet with $\mathcal{L}_{att}$ received notable more votes for perceived similarity compared to those from the model without $\mathcal{L}_{att}$. This finding confirms the effectiveness of the proposed attribute loss in improving the human-perceived similarity of reconstructed faces.

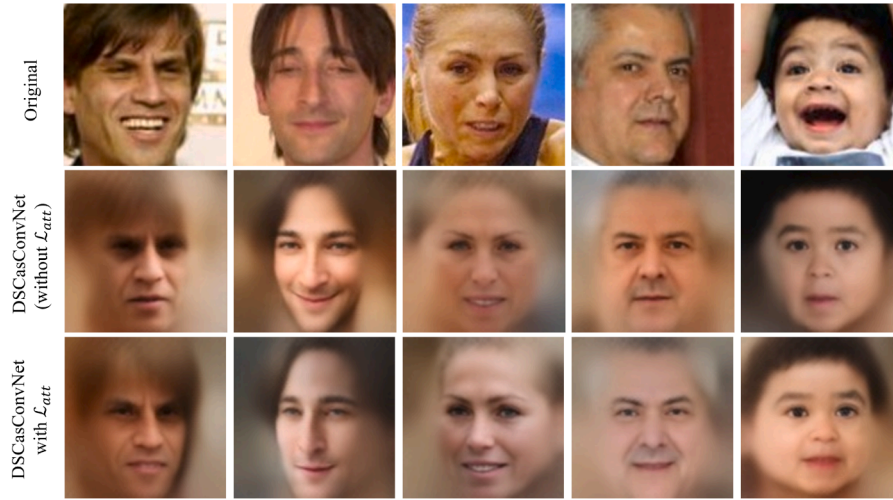


Fig. 17. Visual examples of reconstructed face images, tested on LFW with ArcFace-IResNet100 as the target model and other unseen models.

Table 10

Ablation study of the attribute loss $\mathcal{L}_{att}$ on DSCasConvNet-ArcFace, using LFW as the testing dataset.

Face recognition systems	$\mathcal{L}_{att}$	TARs at FAR = 0.01%		TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑	Type-I↑	Type-II↑
ArcFace (<i>target</i>)	✓	0.9998	0.9623	0.9999	0.9846	1.0000	0.9910
	✗	0.9997	0.9514	0.9998	0.9836	0.9998	0.9908
AdaFace (<i>unseen</i>)	✓	0.8903	0.7582	0.9995	0.9834	0.9998	0.9908
	✗	0.8934	0.7142	0.9995	0.9795	0.9997	0.9903
MagFace (<i>unseen</i>)	✓	0.9980	0.9566	0.9980	0.9566	0.9995	0.9834
	✗	0.9980	0.9347	0.9980	0.9347	0.9995	0.9795
SphereFace (<i>unseen</i>)	✓	0.9261	0.8095	0.9733	0.9264	0.9945	0.9760
	✗	0.9256	0.7483	0.9739	0.9079	0.9938	0.9743
CurricularFace (<i>unseen</i>)	✓	0.9961	0.9557	0.9992	0.9819	0.9996	0.9906
	✗	0.9958	0.9452	0.9986	0.9809	0.9995	0.9898

5.10. Hyperparameter sensitivity analysis

This section analyzes the sensitivity of the proposed method to variations in λ_{att} . To conduct this analysis, we evaluated λ_{att} values of 1, 5, 10, 15, 50, and 100, while fixing all other loss weights at their default value of 1. As illustrated in Fig. 19, our method exhibits considerable robustness to changes in λ_{att} , achieving relatively high performance when $\lambda_{att} = 10$.

5.11. Result stability analysis

To validate the stability of our performance improvements, we conduct multi-seed repeated experiments with error-bar statistics and significance tests. To eliminate training randomness, we train all models five times with distinct random seeds. All results in Table 11 are reported as mean $\pm$ standard deviation across independent runs to demonstrate result stability.

5.12. Time costs and practicality analysis

We evaluated the time costs of UCNNet and the attribute-guided M-StyleGAN3 during both training and testing phases, using ArcFace as the target model. The experiments were conducted on an NVIDIA RTX 3090 GPU. As summarized in Table 12, UCNNet and the attribute-guided M-StyleGAN3 take about 0.04 s and 0.14 s respectively, to reconstruct

a face image, demonstrating their suitability for fast interactive applications. While the training process incurs substantial time costs (52.5 h for UCNNet and 71 h for M-StyleGAN3), this one-time computational overhead is acceptable given the offline nature of model training, thus validating the practical utility of both models.

From a practical perspective, our method only requires black-box access to the target face recognition system, which aligns with realistic attack scenarios. Furthermore, its ability to generalize to unseen face recognition models (AdaFace, MagFace, SphereFace, CurricularFace) further enhances its practical value.

5.13. Discussion

Our experiments present extensive results, from which we would like to draw some conclusions as follows.

- **Attribute guidance boosts inversion effectiveness.** Both human perception and face recognition systems rely heavily on facial attributes (eyes, nose, mouth) for identification. Thus, it is important to reconstruct face images with facial attributes similar to the original. Our experimental results demonstrate that a simple network (UCNNet) can achieve good precision in attack of face recognition systems with the attribute guidance. The performance is also remarkable in the attack of *unseen* face recognition systems. In addition, the integration of attribute guidance into M-StyleGAN3 (Otroshi Shahreza & Marcel,

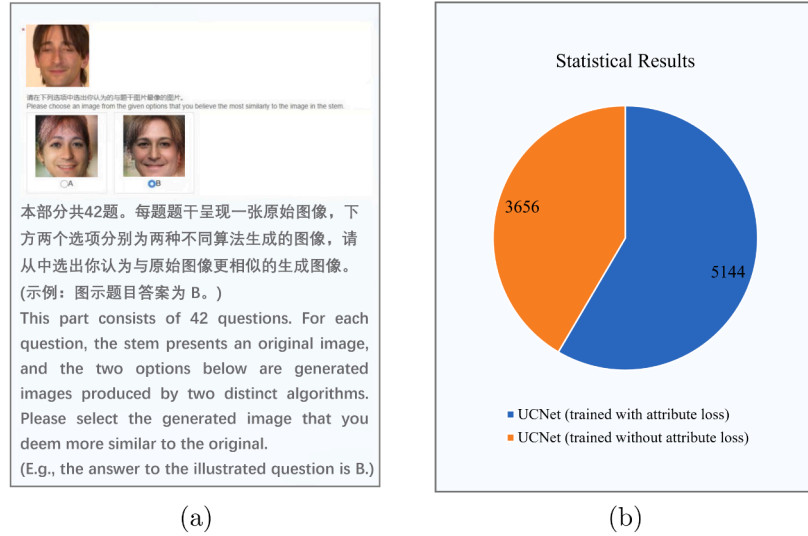


Fig. 18. Questionnaire for evaluating human-perceived similarity: (a) Example of a questionnaire item, and (b) Statistical results.

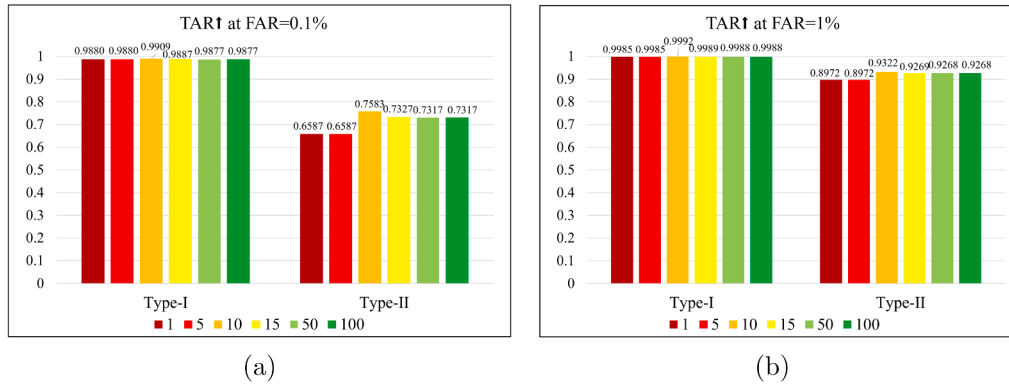


Fig. 19. Performance of attribute-guided M-StyleGAN3 with different λ_{att} , using ArcFace as the target model.

Table 11

TARs (Mean $\pm$ Std) of attribute-guided M-StyleGAN3, tested on LFW with ArcFace as the target model on LFW with 5 random seeds.

Face recognition systems	$\mathcal{L}_{att}$	TARs at FAR = 0.1%		TARs at FAR = 1%	
		Type-I↑	Type-II↑	Type-I↑	Type-II↑
ArcFace (<i>target</i>)	✓	0.9909 $\pm$ 0.0002	0.7569 $\pm$ 0.0016	0.9992 $\pm$ 0.0000	0.9297 $\pm$ 0.0018
	✗	0.9507 $\pm$ 0.0007	0.4653 $\pm$ 0.0017	0.9931 $\pm$ 0.0004	0.8095 $\pm$ 0.0021
AdaFace (<i>unseen</i>)	✓	0.5586 $\pm$ 0.0015	0.2794 $\pm$ 0.0007	0.7764 $\pm$ 0.0005	0.5926 $\pm$ 0.0011
	✗	0.3522 $\pm$ 0.0007	0.0563 $\pm$ 0.0003	0.5863 $\pm$ 0.0021	0.1905 $\pm$ 0.0010
MagFace (<i>unseen</i>)	✓	0.9564 $\pm$ 0.0002	0.7554 $\pm$ 0.0013	0.9929 $\pm$ 0.0001	0.9355 $\pm$ 0.0013
	✗	0.8470 $\pm$ 0.0009	0.3687 $\pm$ 0.0016	0.9670 $\pm$ 0.0006	0.7711 $\pm$ 0.0014
SphereFace (<i>unseen</i>)	✓	0.6216 $\pm$ 0.0007	0.3431 $\pm$ 0.0010	0.8275 $\pm$ 0.0009	0.7037 $\pm$ 0.0011
	✗	0.4163 $\pm$ 0.0008	0.0831 $\pm$ 0.0007	0.6533 $\pm$ 0.0009	0.2801 $\pm$ 0.0013
CurricularFace (<i>unseen</i>)	✓	0.9194 $\pm$ 0.0004	0.6852 $\pm$ 0.0006	0.9801 $\pm$ 0.0005	0.9026 $\pm$ 0.0010
	✗	0.7797 $\pm$ 0.0014	0.3745 $\pm$ 0.0027	0.9237 $\pm$ 0.0003	0.7535 $\pm$ 0.0015

2024) and DSCasConvNet (Shahreza et al., 2024) further improves the performance, verifying the adaptability of our attribute loss.

- It is a good strategy to build the inversion model on the basis of pre-trained GAN models. Many GAN models have been proposed to generate images with high resolution and quality. Several facial template inversion methods are built based on the pre-trained StyleGAN (Dong et al., 2021, 2023; Otrosi Shahreza & Marcel, 2024;

Shahreza & Marcel, 2023b; Vendrow & Vendrow, 2021) and achieve good results. Incorporating our attribute loss into M-StyleGAN3 (Otroshi Shahreza & Marcel, 2024) also improves the performance.

- Reconstructed faces lack sufficient human-perceived similarity to originals. Our attribute loss improves the FAMSE and the precision in the attack of the face recognition system. However, as observed by human vision, the reconstructed face images do not have facial at-

Table 12

The time costs of UCNet and attribute-guided M-StyleGAN3 in training and testing phases, with ArcFace as the target model.

Models	Phase	Number of parameters	Datasets	Image numbers	Image size	Time consumption
UCNet	Training	119,467,011	FFHQ	63,000	128 × 128	52.5 h
	Testing	119,467,011	LFW	13,233	128 × 128	9 min
Attribute-guided M-StyleGAN3	Training	792,064	FFHQ	63,000	128 × 128	71 h
	Testing	14,567,839	LFW	13,233	128 × 128	31 min

Note: The training phase of UCNet consists of 300 epochs, with each epoch comprising 2187 iterations. The training phase of attribute-guided M-StyleGAN3 consists of 20 epochs, with each epoch comprising 15,750 iterations.

tributes similar enough to the originals, as shown in Figs. 11, 13, and 15. More efforts are needed to precisely control these details in image generation.

6. Conclusion

This paper proposes AFTI (Attribute-guided Facial Template Inversion) to address the challenge of facial template inversion by explicitly modeling key facial attributes (eyes, nose, and mouth). We design attribute loss and integrate it into two complementary inversion models: UCNet, a lightweight network, and M-StyleGAN3 that leverages generative capabilities for high-fidelity reconstruction. Both models are enhanced by AFTI: the lightweight design of UCNet underscores the effectiveness of $\mathcal{L}_{att}$, while integrating this loss into pre-trained StyleGAN further improve the performance. Quantitative results demonstrate superior performance across critical metrics, including FAMSE, DFS, and attack precision on face recognition systems. Notably, AFTI achieves remarkable improvements in Type-II attacks and exhibits strong cross-model generalization.

Despite these advancements, there remains room for improvement in the human-perceived similarity between reconstructed images and original faces. How to precisely control face generation, especially the accurate manipulation of facial features, remains a significant challenge.

CRedit authorship contribution statement

Zixuan Shen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Junheng Luo:** Software, Data curation; **Zhihua Xia:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization; **Xi Yang:** Writing – review & editing; **Kaikai Gan:** Validation, Data curation; **Peipeng Yu:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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